

PIN-JIAN CHEN

✉ Email | 🌐 Website | 🆎 ORCID

Education

National Astronomical Observatories, CAS (NAOC)

Beijing

PhD in Astrophysics; Advisor: Xiaodian Chen

Sep 2024 – Expected June 2029

-Research Interests:

Late stage evolution of low- and intermediate-mass stars; Emission-line Nebulae; Pulsating variables and extragalactic distance scale; binary stars; M31&M33; Machine learning's application in astronomy

Department of Astronomy, Yunnan University

Kunming

Bachelor in Astronomy; Advisor: Bingqiu Chen

Sep 2020 – June 2024

-Selected Awards & Honors:

- First Prize Scholarship (Annual), YNU, 2021, 2022, 2023
- Qing-Lai Xiong Astronomy Special Scholarship (Sponsored by YNAO), First Prize (1 recipient), 2021, 2022
- Excellent Graduate of Yunnan Province, 2024; Excellent Bachelor Thesis, YNU, 2024
- Undergraduate Honor Certificate (20 recipients across the university), YNU, 2024

-Programs:

Member of Elite Class for Astronomy

Publications

1. **P.-J. Chen**, B.-Q. Chen*, X.-D. Chen* et al., *A Planetary Nebula from a $5.7 M_{\odot}$ Progenitor in a 90 Myr M31 Star Cluster*, ApJL, 2026, accepted
2. **P.-J. Chen**, B.-Q. Chen*, X.-D. Chen* et al., *Discovery of an Extremely Luminous Type II Cepheid in the Andromeda Giant Stellar Stream: Evidence for a Hierarchical Triple with an Inner Binary Merger*, ApJL, 2026, 997, L24
3. B.-Q. Chen*, G.-X. Li*, ..., **P.-J. Chen**, et al., *A large, long-lived, slowly-expanding superbubble across the Perseus arm*, Nat Commun, 2025, 16, 10558
4. **P.-J. Chen**, B.-Q. Chen*, H.-B. Yuan, et al., *A Massive Yellow Supergiant in the Far Outer Disk of M31: Evidence for in Situ Massive Star Formation beyond the Optical Radius*, AJ, 2025, 170, 89
5. B.-S Zhang, B.-Q. Chen*, H.-B. Yuan, **P.-J. Chen** et al., *Identification of Star Clusters in M31 from PAndAS Images Based on Deep Learning*, ApJS, 2025, 278, 16
6. **P.-J. Chen**, B.-Q. Chen*, X. Fang*, et al., *A Comprehensive Catalog of Emission-line Nebulae, Star Clusters, and Supergiants in M31 from the LAMOST Spectroscopic Survey*, AJ, 2025, 169, 174
7. **P.-J. Chen**, X. Fang*, X.-D. Chen, J.-F. Liu, *Periodic Variability of the Central Stars of Planetary Nebulae Surveyed through the Zwicky Transient Facility*, ApJ, 2025, 980, 227
8. X.-W. Zhang, B.-Q. Chen*, **P.-J. Chen**, et al., *The rotation curve and mass distribution of M31*, MNRAS, 2024, 528, 2653

* indicates corresponding author

Conferences & Workshops

1. Annual Academic Conference of Yunnan Astronomical Society, Puer, Yunnan, China, Sep 16-19, 2023
Contributed talk: Construction of LAMOST-M31 Catalog and Mass Determination of M31, *excellent talk*

Observational Experience

1. 2 nights with Hale 5.1-m Telescope at Palomar Observatory, remote, 2023 (PI: Bingqiu Chen)
2. 3 nights with Hale 5.1-m Telescope at Palomar Observatory, remote, 2024 (PI: Bingqiu Chen)
3. 45 hours with 10.4 m Gran Telescopio Canarias (GTC), remote, 2024-2025 (PI: Rubén García-Benito)
4. 37 hours with 4.2 m William Herschel Telescope (WHT), remote, 2025-2026 (PI: Rubén García-Benito)
5. 2 nights with Cerro Murphy 0.8 m telescope, remote, 2026-2027 (PI: He Zhao)

Skills

Language Abilities: Chinese (Mother tongue), English (Fluent)

Programming Languages: Python, L^AT_EX, HTML, Matlab

Professional Software: IRAF/PyRAF, DS9